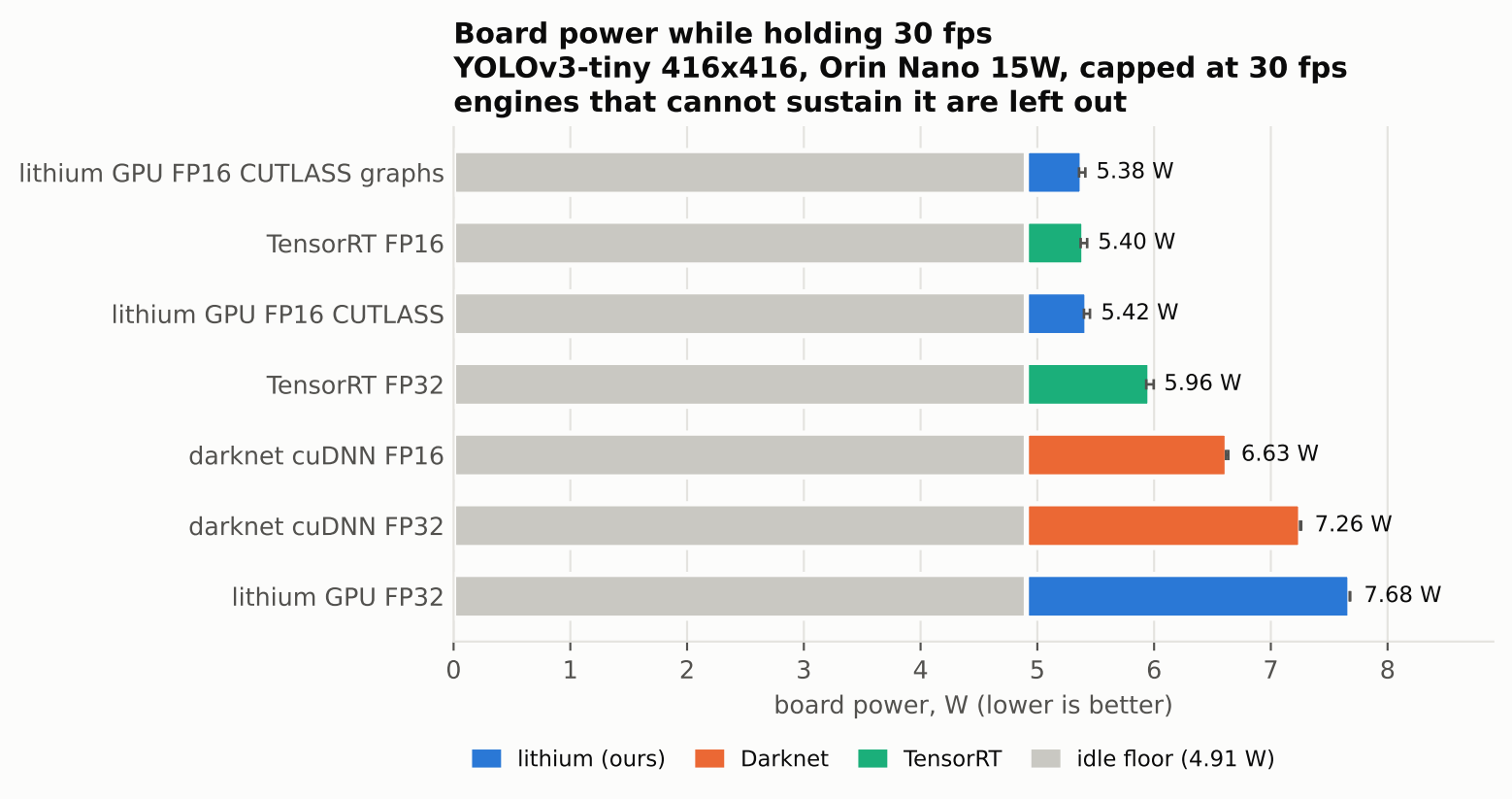
Board power while holding 30 fps
YOLOv3-tiny 416x416, Orin Nano 15W, capped at 30 fps
engines that cannot sustain it are left out
lithium GPU FP16 CUTLASS graphs
TensorRT FP16
lithium GPU FP16 CUTLASS
TensorRT FP32
darknet cuDNN FP16
darknet cuDNN FP32
lithium GPU FP32
5.38 W
5.40 W
5.42 W
5.96 W
6.63 W
7.26 W
7.68 W
0
1
2
3
4
5
6
7
8
board power, W (lower is better)
lithium (ours)
Darknet
TensorRT
idle floor (4.91 W)